

APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

Training loop summary

Algorithm A1 Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors R ,

auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode X with the directed-diffusion GRU to obtain node context H

Compute gate logits Z and soft routing probabilities G

Compute expert forecasts and aggregate them with G

Evaluate L_{pred}

If MoE mode is active:

add L_{bal} on top-K routing statistics

add L_{align} on valid primary regime anchors

if WTB: add L_{force} on MPPT/pitch samples

if WTB: add L_{aux} on wake labels

add L_{smooth} on the final graph snapshot

Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

where Ω_{wake} contains only MPPT and pitch-control samples with defined wake flags. The graph-smoothness penalty is

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}} \sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}} \sum_{i,j} \mathcal{A}_t(i,j) + \epsilon}.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^\perp = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$. A candidate edge activates when

Model-family and supplementary evidence index

TABLE I

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Table A1. In-family model comparison used for mechanism validation.

Model	Routing	Extra terms	Params	Rule
Dense (matched)	Single-dense head	None	WTB 110,012 / ERA5 103,272	Shard-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Router-capacity control
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{\text{bal}} + L_{\text{align}} + L_{\text{force}}$; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Accountability comparator

Information boundary and channel roles

Table A2 makes the leakage and shared-anchor boundary explicit. The routing labels and some gate anchors deliberately share wind-speed and pitch information; the reported NMI/ARI therefore audits compliance with a declared SCADA boundary, not anchor-free discovery.

TABLE II

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Table A2. Information boundary and channel-role audit.

Quantity	Label construction	Model input / gate anchor	Supervised target	Timing boundary
wppd	WTB regime label	history and issue-time anchor	no	$\leq t$ only
pab_mean	WTB regime label	history and issue-time anchor	no	$\leq t$ only
$\text{pabv}_{t-H+1:t}$	no	historical input / issue-time status	no	$\leq t$ only
$\text{pabv}_{t+1:t+P}$	no	no	yes	future target only
Wake score	auxiliary wake flag	graph-derived anchor	no	computed from issue-time graph
Declared regime label	alignment/forcing supervision	no test-time input	no	train/validation supervision only
Confirmed threshold stream	degraded-rule comparator	not a model input	no	delayed/missing/noisy in audit

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top-K load is $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$, giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

For WTB wake supervision,

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

The wake weight is $\tilde{\mathcal{A}}_t(i,j) = \exp(-d_{ij,t}^\parallel / \alpha) \exp(-|d_{ij,t}^\perp| / \beta) \mathbb{I}_{ij,t}^{\text{cone}}$. If wind direction is missing, fall back to $\mathcal{A}^{\text{static}}(i,j) = \exp(-\|\Delta \mathbf{p}_{ij}\|_2 / d_{\text{max}})$. Only the strongest M inbound weights are retained. The wake score and flag are $s_{i,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i,j)$ and

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i,j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph.

Shared constants

TABLE III
***Table A3.** Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

TABLE IV
***Table A4.** Dataset-specific regime thresholds used to construct routing anchors.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of sshf on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

TABLE V
***Table A5.** Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

Statistical claim boundaries

Table A6 separates the train-only RMSE guardrail from paired seed-level tests on the archived full-audit checkpoint. The iTransformer row is the displayed five-seed train-only guardrail gap; the Graph WaveNet and boundary-window rows are legacy full-audit paired/FDR audit rows and should not be read as the current RMSE guardrail.

TABLE VI
***Table A6.** Statistical claim boundaries used for reviewer-facing wording.

Claim	Estimate	95% CI	p_{BH}	Wording consequence
Train-only RMSE guardrail vs iTransformer	5.589	–	–	Displayed guardrail gap for the provenance-corrected rerun; legacy full-audit checkpoint is separate
Legacy full-audit RMSE price vs Graph WaveNet	9.029	–	<0.001	Applies to the archived full-audit checkpoint, not the train-only RMSE guardrail
Legacy boundary-band RMSE price	17.758	–	0.001	Boundary-window price for the archived full-audit checkpoint
Gate NMI vs full physics-aligned MoE	0.040	[0.008, 0.071]	0.066	Positive but not FDR-significant; cite as bounded mechanism contrast
Gate ARI vs full physics-aligned MoE	0.035	[-0.000, 0.078]	0.126	Positive but not FDR-significant; avoid superiority wording
Boundary quantile cost vs GWN physical bin	-0.263M	[-10.704M, 9.782M]	–	CI crosses zero; reserve cost should remain a diagnostic claim
Boundary quantile violation vs GWN physical bin	0.003	[-0.011, 0.020]	–	CI crosses zero; no universal reserve-policy optimality claim

Class-weight sensitivity audit

Table A6b makes the class-weight provenance boundary explicit. The legacy strict-cache row is the originally frozen boundary-router checkpoint used for the headline audit; the train-only rerun recomputes alignment and pitch-forcing class weights from the training split only while preserving the same strict-mask evaluation protocol. The train-only rerun lowers NMI/ARI but remains above the routing-claim threshold.

TABLE VII
***Table A6b.** Class-weight sensitivity audit. Boundary-router rerun after recomputing alignment and pitch-forcing loss weights from the training split only.

Condition	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
Legacy strict-cache weights	236.13	239.86	286.95	0.8716	0.9166
Train-only weight rerun	229.93	231.65	289.63	0.7208	0.7398
Delta	-6.20	-8.21	2.68	-0.1508	-0.1768

Outcome-channel sanity audit

Table A7 adds a bounded check for the shared-anchor concern. It does not use pitch-threshold labels to score the contrast: validation data define a wind-speed-bin power curve, and the test-set comparison is restricted to 9.5–11.5 m s⁻¹ boundary anchors with fine wind-bin adjustment. The result asks whether the recovered gate separates samples with different future active-power response, not whether it discovers a regime without anchors.

TABLE VIII
***Table A7.** Outcome-channel sanity audit for the shared-anchor concern.

Check	Five-seed summary	Interpretation
Boundary cells per seed	3276 MPPT / 10465 pitch	9.5–11.5 m s ⁻¹ test anchors
Future mean power	100.0 ± 17.5 kW	Pitch-gate minus MPPT-gate after wind-bin adjustment
Power-curve residual	100.0 ± 17.5 kW	Validation wind-bin power curve only; no pitch label used
Future power ramp	81.8 ± 24.7 kW	Mean late-horizon minus early-horizon power
Anchor-time Patv	-44.1 ± 32.5 kW	Current active power is not driving the same positive contrast
Residual claim boundary	sanity check only	Mitigates circularity concern; does not prove anchor-free discovery

External-site deployment-gate audit

Table A8 records the go/no-go interpretation used for the external evidence. The first row is the cross-site mechanism replication: retrained on the ENGIE La Haute Borne farm, where blade pitch is directly observed in 99.2% of cells, the routing mechanism recovers the declared boundary and clears the held-out criterion. The remaining rows map the failed or incomplete Kelmarsh/Penmanshiel signals to the deployment action that follows. The table therefore supports a bounded but two-sided claim: the mechanism replicates where the control boundary is observable, while direct gate-bin reserve use at the Kelmarsh/Penmanshiel pair remains a screening protocol rather than an authorized transfer.

TABLE IX
***Table A8.** External-site deployment-gate audit and wording boundary.

Gate	Observed external evidence	Go/no-go rule	Claim consequence
Cross-site mechanism replication (La Haute Borne)	Five-seed chronological routing NMI 0.941, ARI 0.971; pitch observed in 99.2% of cells, no proxy	Held-out NMI ≥ 0.50 with observed pitch after parameters are frozen	Go: anchor-observable mechanism replicates on a second farm; do not claim anchor-free or automatic reserve transfer
Cross-site routing criterion	80/80 runs complete; mean NMI 0.4877 below threshold 0.50; mean ARI 0.5112	Held-out NMI ≥ 0.50 and balanced accuracy ≥ 0.50 after parameters are frozen	No-go for cross-farm router interpretation; cite as negative boundary-condition evidence
Local boundary recalibration	Default test NMI 0.1324 → recalibrated 0.1491 (delta +0.0167)	Rated wind, pitch threshold, boundary band, and gate-map selected on calibration only	Local threshold transfer is insufficient; re-estimate before use
Small-window adaptation	40/40 routing runs adapted; chronological balanced accuracy 0.4787 below 0.50	Small calibration windows must still pass the frozen held-out routing criterion	Calibration alone does not authorize external reserve use
Sensor and boundary support	Penmanshiel-to-Kelmarsh leave-one pitch-feature coverage 0.0000 and effective boundary cells 0	Pre-declared calibration window with enough boundary cells, active power, availability mask, and pitch/proxy overlap	No physical-router interpretation without observability
External reserve-use decision	Upstream observability and held-out routing gates fail before external reserve allocation	Evaluate reserve only after observability and held-out routing gates pass	Withhold gate-bin reserve use outside WTB; report a deployment protocol only

La Haute Borne anchor-observability replay audit

Table A9 audits the load-bearing channels behind the La Haute Borne cross-site mechanism replication. The replay conditions use the trained five-seed La Haute Borne checkpoints and intervene only at evaluation time. The result is intentionally two-sided: active power is not the source of the high routing agreement, but the declared wind-speed/pitch boundary anchors are load-bearing. This supports citing La Haute Borne as anchor-observable cross-site mechanism replication and blocks any anchor-free discovery or automatic cross-site reserve-use wording.

TABLE X
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Table A9. La Haute Borne anchor-observability replay audit.

Replay condition	NMI	Δ NMI	Interpretation
Actual replay	0.941	0.000	reference five-seed route
Zero pitch anchor	0.696	0.245	partial alignment remains
Zero wind-speed anchor	0.752	0.189	partial alignment remains
Zero active-power anchor	0.953	-0.013	active power is not load-bearing
Zero wind+pitch anchors	0.001	0.940	boundary alignment collapses
Randomize anchor physics	0.028	0.913	physical anchor mapping collapses
Wind-speed only	0.752	0.189	single-anchor partial control
Pitch only	0.670	0.271	single-anchor partial control

Early-warning detection consequence

Table A10 reports the reviewer-facing detector control for the label-degradation audit. The simple classifier is a validation-fit multinomial logistic regression on the same issue-time anchors. It matches or exceeds the gate on clean-anchor standalone detection, so the manuscript claims auditable in-model route attribution rather than classifier superiority. Recovered cells are turbine-time cells per seed inside the six-step MPPT-to-pitch window and are not MWh, currency, or dispatch-cost estimates.

TABLE XI
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Table A10. Early-warning detector control under degraded threshold labels.

Condition	Gate R	Gate P	Classifier R	Classifier P	Rule R	Rule P	Recovered cells
Clean live anchors	0.960	0.759	1.000	0.879	1.000	0.879	–
Delay, 6 steps	0.960	0.759	1.000	0.879	0.196	0.342	565.6 $\pm$ 20.6
50% label availability	0.960	0.759	1.000	0.879	0.508	0.877	334.6 $\pm$ 28.1
Sensor noise, strongest	0.960	0.759	0.885	0.758	0.652	0.804	227.8 $\pm$ 20.9

R/P denote recall and precision on early pitch-window cells. The simple classifier is a validation-fit logistic model on the same issue-time anchors; it is a detector control, not a routed forecaster. In sensor-noise rows, gate values are the saved clean-route audit; classifier/rule values are recomputed from noisy anchors.

Gate route-evolution diagnostic

Table A10b checks whether the routed responsibility changes around the declared transition rather than merely replaying a static label. Matching to the new regime peaks at the transition step and drops under lead/lag shifts, supporting a route-evolution audit while not replacing the modular classifier control.

TABLE XII
***Table A10b.** Gate route-evolution diagnostic around MPPT-to-pitch transitions.

Lag step	Match rate	SD	Seeds
-6	0.405	0.005	5
-3	0.361	0.010	5
+0	0.815	0.059	5
+3	0.633	0.002	5
+6	0.610	0.002	5

TABLE XIII
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Table A10c. Modular classifier reserve control and responsibility-chain boundary.

Control	Result and claim boundary
GWN+classifier-bin	86.54M improves over GWN/global (88.80M) but trails physical-bin/gate-bin (84.31M/84.58M); paired modular-vs-gate CI crosses zero, so this is a tested modular control, not an in-model route-responsibility replacement.

Reserve-policy claim-boundary audit

Table A11 is the compact reviewer-facing boundary audit. The same-router boundary comparison has seed-paired uncertainty support, while physical-bin, full-sample, and cross-backbone comparisons still bound the claim away from policy optimality or market-dispatch value.

TABLE XIV
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Table A11. Reserve-policy claim-boundary audit.

Boundary	Key evidence	Limit
Same-model boundary reserve effect	$\rho = 10$: Δ cost -3.55M [CI -4.65M,-2.45M]; Δ viol. -0.0138 [CI -0.0203,-0.0072]; shortage -0.54M [CI -0.78M,-0.30M]	Same-model diagnostic; not cross-model optimal
Physical-bin quantile comparator	Boundary phys. 83.78M/0.0880; GWN phys. 84.31M/0.0931; gate 84.58M/0.0900	Physical bins remain competitive
Cost-ratio applicability	Active at $\rho = 5$ –10; narrows at 20; $\rho = 50$ favors global (+5.93M, +0.0035)	Moderate-cost window only
Full-sample system value	GWN/global 464.07M/0.0901; gate-bin 481.36M/0.1214	No system-wide dispatch claim
Cross-backbone/full-sample uncertainty	Δ cost +17.28M; 95% CI [-52.09M,+87.95M]; $p=0.752$	Not system-wide or cross-backbone
Operational scope	Validation-frozen shortfall quantiles; no OPF, unit commitment, market clearing, or prices	Screening audit only

Same-model intervals are $n = 5$ seed-paired bootstrap mean CIs for gate-bin minus same-router global at $\rho = 10$.

Engineering-unit reserve-value translation

Table A12 translates the reserve audit into MWh-equivalent forecast-cell accounting and an illustrative 100 EUR/MWh reserve-cost-scale marker; it is not a market-settlement, OPF, unit-commitment, or security-constrained dispatch result.

TABLE XV
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Table A12. Engineering-unit reserve-value translation at $\rho = 10$.

Comparison	Δ reserve	Avoided shortage	Δ cost	Cost-scale@100
Boundary gate-bin vs same-router global	+1846.9	+539.8	-3551.4	-355k
Boundary gate-bin vs GWN physical-bin	+2319.2	+205.6	+262.9	+26k
Boundary gate-bin vs GWN global full sample	+3148.2	-1413.5	+17283.2	+1728k

Values are MWh-equivalent forecast-cell accounting with $\Delta t = 1/6$ h.

Cost-scale@100 is an illustrative reserve-cost scale marker at 100 EUR/MWh, not market revenue, settlement value, OPF, or unit-commitment output. Use same-router/global as the bounded boundary-window diagnostic; physical-bin and full-sample rows block reserve superiority.